Sahaj Bhadja

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Summary

Software engineer with 2+ years building **C++/C#/Python systems**, **real-time simulation and UI tooling**, and **testable gameplay-style features** with strong debugging, iteration speed, and cross-functional collaboration.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **LA, California**
Software Engineer *September 2025 - Present*
- Built **interactive simulation and analysis tools** for 500+ users, reducing iteration latency from **1.2s to 350ms** through low-level optimization and caching.
- Implemented **test-case driven validation workflows** for 5K+ monthly runs, improving feature stability and accelerating debug turnaround for fast-release cycles.

Grad8 **LA, California**
Software Engineer Intern *June 2025 - August 2025*
- Developed **user-facing product features** handling 10K+ daily interactions, translating design requirements into responsive UI components and reliable integration logic.
- Improved **release quality** with CI/CD checks and structured defect triage, reducing issue turnaround by **60%** across cross-functional sprint delivery.

USC AD Labs **Los Angeles, CA**
ML Research Assistant *January 2025 - May 2025*
- Built **trajectory simulation pipelines** over 100K+ records for reproducible behavior analysis, enabling robust tuning of control and decision logic.
- Produced **system documentation and test scenarios** pre-release, improving handoff quality and reducing ambiguity during implementation and QA cycles.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 - July 2024*
- Engineered **C++/Python-integrated service modules** with AWS workflows, improving response times by **30%** while maintaining reliability under production load.
- Maintained **20+ automated test suites** with Jenkins, PyTest, and Mockito, improving build confidence by **68%** and reducing regression defects.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 - August 2023*
- Developed **real-time UI modules** using MERN + WebSocket architecture for 2M+ synchronized records, improving responsiveness of event-driven interfaces.
- Automated **Docker-based build and test flows** with Jenkins, reducing release time by **50%** and improving deployment consistency.

Skills

Languages: **C++**, **C#**, Python, JavaScript/TypeScript, SQL, Bash
Gameplay & UI: real-time simulation logic, interaction tooling, state-driven UI flows, debugging/performance tuning
Systems: Linux/UNIX workflows, distributed integrations, API services, event-driven processing
Quality: test-case design, CI/CD pipelines, defect triage with QA, reproducible validation workflows
Tooling: Git/GitHub, Jenkins, Docker, Agile collaboration, technical documentation

Projects

CPU Scheduling Simulator Code | *C++, React.js, JavaScript*
- Implemented **FCFS, SJF, SRT, RR, and MLQ algorithms** with interactive visualizations and metrics tooling, reducing debugging time by **20%** for scheduler behavior analysis.

Tactical Arena Prototype | *C#, Unity, ScriptableObjects*
- Built **core gameplay systems** including input-driven combat loops, cooldown/state handling, and HUD integration, improving feature iteration speed with modular component architecture.

Linguistix Code | *Python, AWS, Docker, CI/CD*
- Implemented a **containerized service pipeline** with automated CI/CD validation, strengthening release reliability and reproducibility across iterative product changes.